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Tipcart

Abstract

A tipcart, comprising a frame, a dump box hinged on the frame, and a first pin shaft for connecting the frame and the dump box, a front wheel bracket and a rear wheel bracket are respectively fixedly connected under a front and rear ends of the frame, both ends of the front wheel bracket and the rear wheel bracket are respectively provided with a front wheel mounting shaft and a rear wheel mounting shaft, and wheels are mounted on both the front wheel mounting shaft and the rear wheel mounting shaft. The first pin shaft is arranged at the rear end of the frame, and the first pin shaft is located between the front wheel mounting shaft and the rear wheel mounting shaft.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. continuation of co-pending International Patent Application No. PCT/CN2021/110198, filed on Aug. 3, 2021, which claims the benefit and priority of Chinese Patent Application Number 202121307491.6, filed on Jun. 11, 2021 with China National Intellectual Property Administration, the disclosures of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

(2) The present application relates to the technical field of tipcars, in particular to a novel tipcart.

BACKGROUND

(3) Tipcart is a kind of vehicle for short-distance transportation of materials with tippable dump box. Due to its low cost, simple maintenance, convenient operation and wide application. It is suitable for short-distance transportation of sand, earth, coal, ore or other various bulk materials. It is a very practical transport device during the daily handling of objects. Most of the existing Tipcars use a hinge shaft to connect the dump box to the frame. Since the dump box is usually made of a plastic-like integral molding material, the connection point with the hinge shaft is usually not resistant to wear. This leads to serious wear and tear of the connection points after frequent use of the dumping and dumping function, thus increasing the frequency of replacement of the dump box, and most of the existing tipcars are designed be pushed by hand, which is inconvenient to use a trailer towing when heavier cargo is loaded in the cart.

(4) Therefore, the existing technology still needs to be further developed and improved.

SUMMARY

(5) In view of various deficiencies of the prior art, to solve the above problems, a novel tipcart is proposed. The present application provides the following technical solution:

(6) A tipcart, comprising a frame, a dump box hinged on the frame, and a first pin shaft for connecting the frame and the dump box, a front wheel bracket and a rear wheel bracket are respectively fixedly connected under a front and rear ends of the frame, both ends of the front wheel bracket and the rear wheel bracket are respectively provided with a front wheel mounting shaft and a rear wheel mounting shaft, and wheels are mounted on both the front wheel mounting shaft and the rear wheel mounting shaft, the first pin shaft is arranged at the rear end of the frame, and the first pin shaft is located between the front wheel mounting shaft and the rear wheel mounting shaft.

(7) Furthermore, the front wheel bracket and the rear wheel bracket are both U-shaped brackets, wherein the end of the rear wheel bracket is horizontally extended backward to form an extension frame for mounting the rear wheel mounting shaft.

(8) Furthermore, the first pin shaft is arranged between the rear wheel bracket and the rear wheel mounting shaft.

(9) Furthermore, a connecting plate is fixedly arranged under the rear end of the dump box, and left and right ends of the connecting plate extend downward to form a hinge plate, wherein the hinge plate is provided with a hinge hole, and the first pin shaft passes through the hinge hole to hinge the connecting plate and the frame, furthermore, the hinge hole is arranged in a middle position of the hinge plate.

(10) Furthermore, the connecting plate is a frame structure with a hollow center.

(11) Furthermore, a reinforcing rod is also fixedly connected between the rear wheel bracket and the frame, and the reinforcing rod, the rear wheel bracket and the frame are jointly enclosed to form a triangular support structure.

(12) Furthermore, a handle for pushing and pulling the tipcart is also fixedly connected to the frame, and the handle comprises a connecting rod fixedly connected to the frame, an extension rod hinged on the end of the connecting rod, and a hand pull ring connected to the extension rod.

(13) Furthermore, a positioning hole is provided under the hand pull ring, a mounting hole is

provided on the extension rod, after a second pin shaft passes through the mounting hole and the positioning hole, the second pin shaft detachably fixes the hand pull ring on the extension rod, two sets of the mounting holes are arranged on the extension rod at intervals, and a distance between the two sets of mounting holes is not less than the height of the hand pull ring, the bottom of the hand pull ring is fixedly connected with a limiting rod, and a limiting hole is provided in the corresponding extension rod, and the limiting rod extends into the limiting hole so that the hand pull ring is slidably connected to the extension rod.

(14) Furthermore, the lower part of the hand pull ring is rotatably connected with an extension rod, and the end of the extension rod is provided with a connection hole for connecting a trailer.

(15) Furthermore, the front end of the dump box is provided with a handle for easily dumping the dump box, and a clip for clipping the dump box on the frame is recessed inwardly under the handle, a clip ring for connecting the clip is provided on the frame, the clip and the clip ring are fixedly connected in a detachable manner, and the handle and the dump box are hingedly connected by a third pin shaft, and a return spring for maintaining the handle in the clipping position is sleeved on the third pin shaft.

Beneficial Effects

(16) 1. By adding a connecting plate to the original tipcart, a connecting point between the connecting plate and the dump box is additionally provided, thereby reducing the friction strength of the hinge point at the dump box. The connecting plate is made of high-strength metal plate, which effectively extends the service life of the dump box; 2. By arranging the hinge hole in the middle position of the hinge plate, the connecting plate rotates with the rotation of the dump box, which is convenient for the dumping process to keep the dump box at least in a vertical state, which is convenient for dumping the goods in the dump box completely. It is helpful for the frame to reinforce and support the dump box extending out of the support range of the frame, further reducing the impact of the goods squeezing the dump box, and extending the service life of the dump box; 3. By arranging the hinged dump box and the first pin shaft of the frame between the front wheel mounting shaft and the rear wheel mounting shaft, the center of gravity of the tipcart is located between the two wheels when loading goods, so as to avoid overturning caused by uneven loading of goods in the dump box during unloading.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic structural diagram of a novel tipcart according to a first embodiment of the present application;

(2) FIG. 2 is a front view schematic structural diagram of a novel tipcart according to a first embodiment of the present application;

(3) FIG. 3 is a schematic structural diagram of a novel tipcart according to the first embodiment of the present application after removing the dump box, wheels and handles;

(4) FIG. 4 is a front view schematic structural diagram of a novel tipcart according to an embodiment of the present application after removing the dump box, wheels and handles;

(5) FIG. 5 is a top view schematic structural diagram of a novel tipcart according to an embodiment of the present application after removing the dump box, wheels and handles;

(6) FIG. 6 is a schematic structural diagram of the handle according to an embodiment of the present application

REFERENCE SIGNS

(7) **100**, Frame; **110**, Front wheel bracket; **111**, Front wheel mounting shaft; **120**, Rear wheel bracket; **121**, Rear wheel mounting shaft; **122**, Extension frame; **123**, Reinforcing rod; **130**, Connecting rod; **140**, Extension Rod; **150**, Hand pull ring; **160**, Mounting hole; **170**, Limiting rod;

180, Connecting hole; **190**, First pin shaft; **200**, Dump box; **210**, Handle; **211**, Clip; **212**, return spring; **220**, Clip ring; **300**, Wheel; **400**, Connecting plate.

DETAILED DESCRIPTION

(8) In order to make those skilled in the art better understand the technical solutions of the present application, the technical solutions of the present application will be described clearly and completely below with reference to the accompanying drawings of the present application. Based on the embodiments in the present application, other similar embodiments obtained by those of ordinary skill in the art without creative work shall fall within the protection scope of the present application. In addition, the directional terms mentioned in the following embodiments, such as “up”, “down”, “left”, “right”, etc., are only the directions with reference to the drawings. Therefore, the directional terms used are used to illustrate rather than limit the present application.

Embodiment 1

(9) As shown in FIG. 1-5, a novel tipcart comprises a frame **100**, a dump box **200** hinged on the frame **100**, and a first pin shaft **190** for connecting the frame **100** and the dump box **200**, a front wheel bracket **110** and a rear wheel bracket **120** are respectively fixedly connected under a front and rear ends of the frame **100**, both ends of the front wheel bracket **110** and the rear wheel bracket **120** are respectively provided with a front wheel mounting shaft **111** and a rear wheel mounting shaft **121**, and wheels **300** are mounted on both the front wheel mounting shaft **111** and the rear wheel mounting shaft **121**, the first pin shaft **190** is arranged at the rear end of the frame **100**, and the first pin shaft **190** is located between the front wheel mounting shaft **111** and the rear wheel mounting shaft **121**. By arranging the hinged dump box **200** and the first pin shaft **190** of the frame **100** between the front wheel mounting shaft **111** and the rear wheel mounting shaft **121**, the center of gravity of the tipcart is located between the two wheels **300** when loading goods, so as to avoid overturning caused by uneven loading of goods in the dump box **200** during unloading.

(10) Furthermore, the front wheel bracket **110** and the rear wheel bracket **120** are both U-shaped brackets, wherein the end of the rear wheel bracket **120** is horizontally extended backward to form an extension frame **122** for mounting the rear wheel mounting shaft **121**. The vertical U-shaped bracket can avoid interference with the wheels **300** or the frame **100** during the dumping process of the dump box **200**. A triangle plate for strengthening the connection strength is also fixedly connected between the extension frame **122** and the rear wheel bracket **120**, so as to strengthen the connection strength between the extension frame **122** and the rear wheel bracket **120**.

(11) Furthermore, the first pin shaft **190** is arranged between the rear wheel bracket **120** and the rear wheel mounting shaft **121**. The length of the extension frame **122** is as small as possible, but the position of the first pin shaft **190** should be located between the rear wheel bracket **120** and the rear wheel mounting shaft **121**, wherein during the dumping of the dump box **200**, the gravity of which moves down into a triangular area formed by the rear wheel bracket **120**, the rear wheel mounting shaft **121** and the first pin shaft **190**, so as to share the pressure on the first pin shaft **190** and extend the service life of the first pin shaft **190**.

(12) Furthermore, a connecting plate **400** is fixedly arranged under the rear end of the dump box **200**, and left and right ends of the connecting plate **400** extend downward to form a hinge plate, wherein the hinge plate is provided with a hinge hole, and the first pin shaft **190** passes through the hinge hole to hinge the connecting plate **400** and the frame **100**. By adding a connecting plate **400** to the original tipcart, a connecting point between the connecting plate **400** and the dump box **200** is additionally provided, thereby reducing the friction strength of the hinge point at the dump box **200**. The connecting plate **400** is made of high-strength metal plate, which effectively extends the service life of the dump box **200**.

(13) Furthermore, the hinge hole is arranged in a middle position of the hinge plate. The rear end of the dump box **200** is arranged to protrude from the frame **100**. By arranging the hinge hole in the middle position of the hinge plate, the connecting plate **400** rotates with the rotation of the dump box **200**, which is convenient for the dumping process to keep the dump box **200** at least in a

vertical state, which is convenient for dumping the goods in the dump box **200** completely. It is helpful for the frame **100** to reinforce and support the dump box **200** extending out of the support range of the frame **100**, further reducing the impact of the goods squeezing the dump box **200**, and extending the service life of the dump box **200**.

(14) Furthermore, the connecting plate **400** is a frame structure with a hollow center. The frame structure is light in weight, and can well complete the supporting and fixing function.

(15) Furthermore, the wheel bracket comprises a front wheel bracket **110** and a rear wheel bracket **120**, a reinforcing rod **123** is also fixedly connected between the rear wheel bracket **120** and the frame **100**, and the reinforcing rod **123**, the rear wheel bracket **120** and the frame **100** are jointly enclosed to form a triangular support structure. Since the frame **100** and the wheel bracket are only fastened and connected by fasteners such as bolts, and the rear wheel bracket **120** also serves as a turning support point, loosening may occur due to frequent sudden force during long-term dumping process. The rear wheel bracket **120** and the vehicle frame **100** are connected triangularly by adding a reinforcing rod **123**, which improves the connection strength and reduces the possibility of looseness.

(16) Furthermore, a handle for pushing and pulling the tipcart is also fixedly connected to the frame **100**, and the handle comprises a connecting rod **120** fixedly connected to the frame **100**, an extension rod **140** hinged on the end of the connecting rod **130**, and a hand pull ring **150** connected to the extension rod **140**. The connecting rod **130** is horizontally fixed on the side of the front end of the frame **100** or the side of the front wheel bracket **110**. The other end of the connecting rod **130** is hinged with one end of the extension rod **140**. The extension rod **140** is a V-shaped rod, which adapts to the push and pulls of people with different heights, and is convenient for storage at the same time. The other end of the extension rod **140** is detachably and fixedly connected with a hand pull ring **150**, which facilitates grasping and improves the comfort of personnel when pushing and pulling.

(17) Furthermore, a positioning hole is provided under the hand pull ring **150**, a mounting hole **160** is provided on the extension rod **140**, after a second pin shaft passes through the mounting hole **160** and the positioning hole, the second pin shaft detachably fixes the hand pull ring **150** on the extension rod **140**.

(18) Furthermore, two sets of the mounting holes **160** are arranged on the extension rod **140** at intervals, and a distance between the two sets of mounting holes **160** is not less than the height of the hand pull ring **150**. Each set of mounting holes **160** includes two mounting holes **160** disposed opposite to both sides of the extension rod **140**. When the hand pull ring **150** is fixed in the upper mounting hole **160** through the second pin shaft, the hand pull ring **150** is located at the end of the extension rod **140** and higher than the extension rod **140**, which is convenient for manual pushing and pulling. When the hand pull ring **150** is fixed in the lower mounting hole **160** through the second pin shaft, the hand pull ring **150** is located inside the extension rod **140**, and the upper mounting hole **160** is higher than the highest point of the hand pull ring **150**, which is convenient to hook onto a trailer for pushing and pulling. Therefore, by arranging two sets of mounting holes **160** and a detachable fixed hand pull ring **150** at the end of the extension rod **140**, the applicable scope of the push-pull method of the tipcart is increased.

(19) Furthermore, the bottom of the hand pull ring **150** is fixedly connected with a limiting rod **170**, and a limiting hole is provided in the corresponding extension rod **140**, and the limiting rod **170** extends into the limiting hole so that the hand pull ring **150** is slidably connected to the extension rod **140**. By arranging the slidably connected limit rod **170** and the limit hole, it is convenient to quickly change the position of the hand pull ring **150**, and the installation efficiency is improved.

(20) Furthermore, the front end of the dump box **200** is provided with a handle **210** for easily dumping the dump box **200**, and an clip **211** for clipping the dump box **200** on the frame **100** is recessed inwardly under the handle **210**, a clip ring **220** for connecting the clip **211** is provided on the frame **100**, the clip **211** and the clip ring **220** are fixedly connected in a detachable manner, and

the handle **210** and the dump box **200** are hingedly connected by a third pin shaft, and a return spring **212** for maintaining the handle **210** in a clipping position is sleeved on the third pin shaft.

Embodiment 2

(21) The structure of the tipcart of this embodiment is basically the same as that of the previous embodiment, and the difference is only in that: as shown in the figure, the lower part of the handle ring **150** is rotatably connected with the extension rod **140**, and the end of the extension rod **140** is provided with a connecting hole **180** for connecting a trailer. When the hand pull ring **150** is rotated to the same horizontal plane as the extension direction of the extension rod **140**, the connecting hole **180** is located inside the hand pull ring **150**, which is convenient for manually pulling the hand pull ring **150** to operate the tipcart. When the hand pull ring **150** is rotated to a position close to the extension rod **140**, the connecting hole **180** is exposed at this time, which is convenient for hooking the trailer for pushing and pulling. Therefore, by providing the rotatable hand pull ring **150** and the connecting hole **180** at the end of the extension rod **140**, the applicable range of the push-pull method of the tipcart is improved.

(22) The application has been described in detail above. The above is only the preferred embodiment of the application, and should not limit the scope of implementation of the application, that is, all equivalent changes and modifications made according to the scope of the application should still belong to the scope of the application.

Claims

1. A tipcart, comprising a frame, a dump box hinged on the frame, and a first pin shaft for connecting the frame and the dump box, a front wheel bracket and a rear wheel bracket are respectively fixedly connected under a front and rear ends of the frame, both ends of the front wheel bracket and the rear wheel bracket are respectively provided with a front wheel mounting shaft and a rear wheel mounting shaft, and wheels are mounted on both the front wheel mounting shaft and the rear wheel mounting shaft, the first pin shaft is arranged at a rear end of the frame, and the first pin shaft is located between the front wheel mounting shaft and the rear wheel mounting shaft; wherein an end of the rear wheel bracket is horizontally extended backward to form an extension frame for mounting the rear wheel mounting shaft, the first pin shaft is arranged between the rear wheel bracket and the rear wheel mounting shaft, wherein during a dumping of the dump box, the gravity of which moves down into a triangular area formed by the rear wheel bracket, the rear wheel mounting shaft and the first pin shaft; wherein a connecting plate is fixedly arranged under a rear end of the dump box, and left and right ends of the connecting plate extend downward to form a hinge plate, wherein the hinge plate is provided with a hinge hole, and the first pin shaft passes through the hinge hole to hinge the connecting plate and the frame, and the connecting plate rotates with a rotation of the dump box; wherein the front wheel bracket and the rear wheel bracket are both U-shaped brackets.
2. The tipcart according to claim 1, wherein the hinge hole is arranged in a middle position of the hinge plate.
3. The tipcart according to claim 2, wherein the connecting plate is a frame structure with a hollow center.
4. The tipcart according to claim 1, wherein a reinforcing rod is also fixedly connected between the rear wheel bracket and the frame, and the reinforcing rod, the rear wheel bracket and the frame are jointly enclosed to form a triangular support structure.
5. The tipcart according to claim 1, wherein a handle for pushing and pulling the tipcart is also fixedly connected to the frame, and the handle comprises a connecting rod fixedly connected to the frame, an extension rod hinged on an end of the connecting rod, and a hand pull ring connected to the extension rod.
6. The tipcart according to claim 5, wherein a positioning hole is provided under the hand pull ring,

a mounting hole is provided on the extension rod, after a second pin shaft passes through the mounting hole and the positioning hole, the second pin shaft detachably fixes the hand pull ring on the extension rod, two sets of the mounting holes are arranged on the extension rod at intervals, and a distance between the two sets of mounting holes is not less than a height of the hand pull ring, a bottom of the hand pull ring is fixedly connected with a limiting rod, and a limiting hole is provided in the corresponding extension rod, and the limiting rod extends into the limiting hole so that the hand pull ring is slidably connected to the extension rod.

7. The tipcart according to claim 5, wherein a lower part of the hand pull ring is rotatably connected with an extension rod, and an end of the extension rod is provided with a connection hole for connecting a trailer.

8. The tipcart according to claim 1, wherein a front end of the dump box is provided with a handle for easily dumping the dump box, and a clip for clipping the dump box on the frame is recessed inwardly under the handle, a clip ring for connecting the clip is provided on the frame, the clip and the clip ring are fixedly connected in a detachable manner, and the handle and the dump box are hingedly connected by a third pin shaft, and a return spring for maintaining the handle in a clipping position is sleeved on the third pin shaft.
